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1. MATLAB Source File

EEL4102_Audio_GUI.m

Audio file used Video Project 1 - Copy.wav

2. Written Report Draft

GUI Design Explanation

This project implements a MATLAB graphical user interface for audio signal processing. The GUI allows the user to load a `.wav` audio file, select one of five processing modes, apply the selected effect, play the original or processed audio, pause or stop playback, and compare the original and processed signals using Power Spectral Density plots.

The GUI is organized into three main sections. The left side contains the controls, including the audio loading button, processing mode dropdown menu, filter parameter inputs, and playback buttons. The middle plot displays the PSD of the original audio signal, while the right plot displays the PSD of the processed signal. This layout makes it easy to compare the input and output visually and understand how each processing mode changes the signal.

Processing Mode Implementation

The Cave, Cathedral, and Stadium modes are implemented using convolution. In these modes, the input audio is convolved with a synthetic impulse response that represents the acoustic behavior of each space. The impulse responses are generated using exponentially decaying noise and delayed reflection peaks. The Cathedral mode uses a longer decay time, which creates a larger and more spacious reverb effect. The Cave and Stadium modes use different reflection delays and decay values to simulate different environments.

The Karaoke and Off-Vocal modes are implemented using `lsim()`. Instead of using convolution with a room impulse response, these modes simulate the audio passing through a linear filter system. Karaoke mode uses a notch-style filter to reduce the selected vocal frequency range, usually around 300 Hz to 3400 Hz. Off-Vocal mode uses a band-pass filter to emphasize the vocal range while reducing frequencies outside that range.

Discussion of Results

For the Cave, Cathedral, and Stadium modes, the processed audio sounds more spacious because convolution adds echoes and reflections. The PSD of the processed signal usually keeps a similar general frequency range compared to the original signal, but the power distribution may change because the reverb spreads signal energy over time. The Cathedral mode usually creates the longest echo tail and strongest reverb effect.

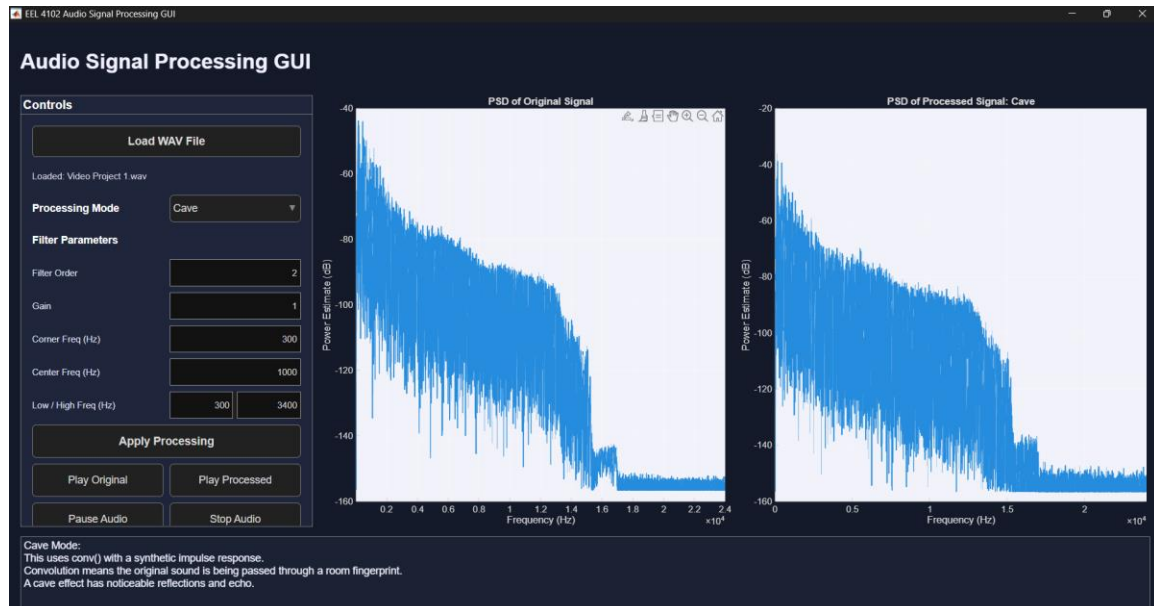
For Karaoke mode, the processed signal should show reduced power in the selected vocal frequency band. This happens because the notch filter weakens frequencies commonly associated with vocals. The vocal removal may not be perfect because vocals overlap with instruments, but the PSD should still show a reduction around the chosen frequency range.

For Off-Vocal mode, the processed signal emphasizes the selected vocal band. The output may sound thinner because frequencies outside the vocal range are reduced. In the PSD plot, the processed signal should show more concentration around the selected band and less energy outside that range.

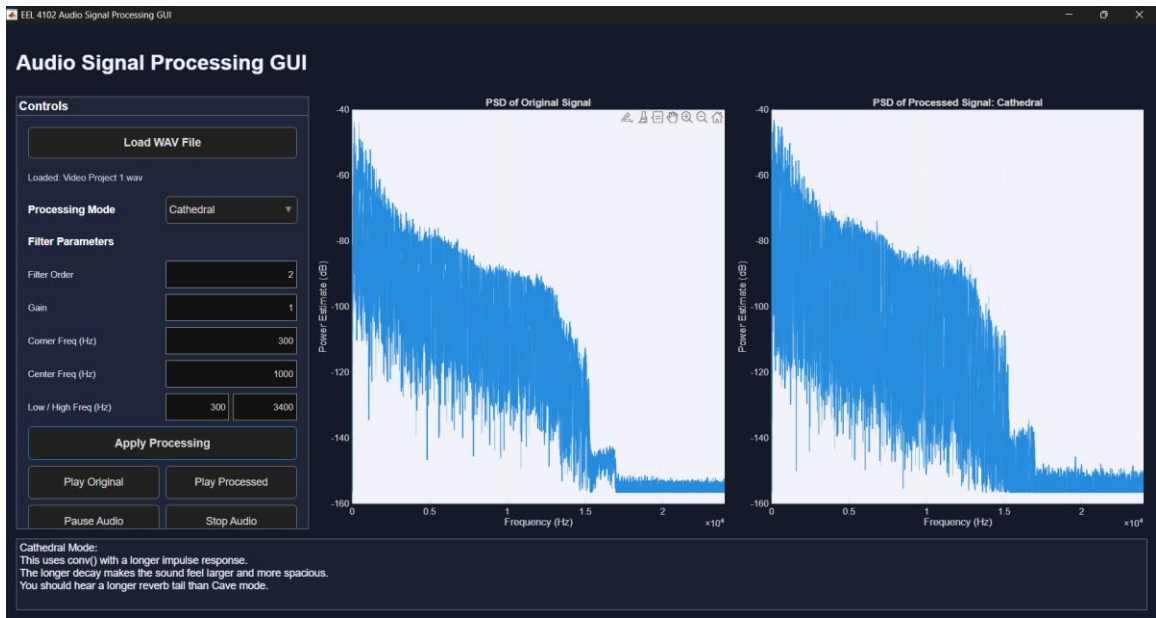
Overall, the PSD comparison is useful because it shows how each processing mode changes the frequency content of the signal. The room effects mainly change the time behavior of the audio by adding reflections, while the Karaoke and Off-Vocal modes directly change the frequency content using filters.

3. Screenshots

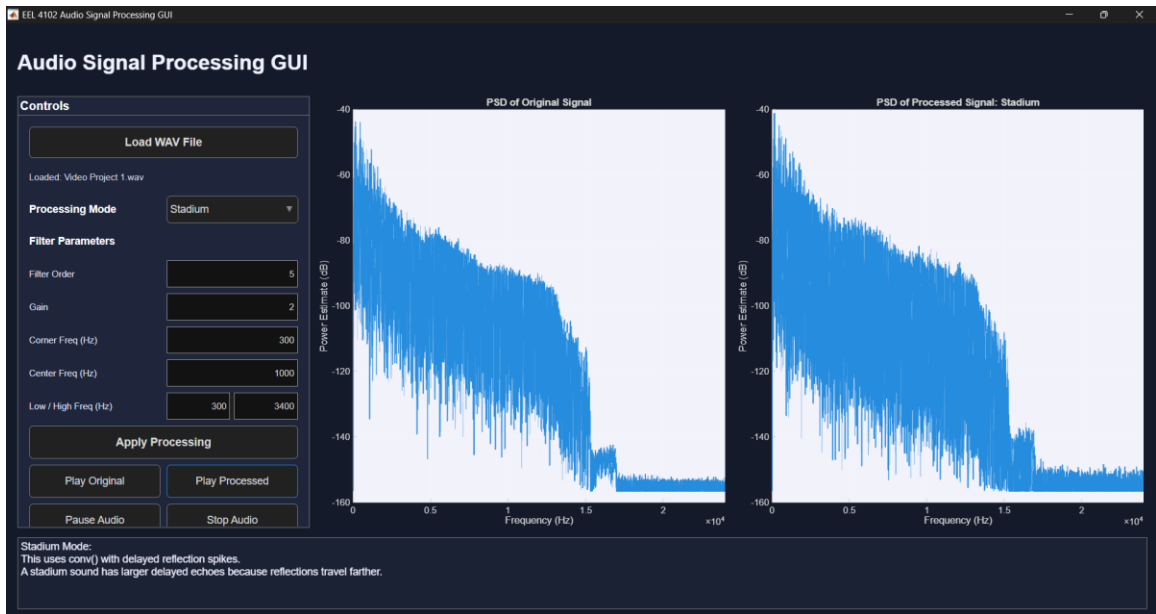
3.1 Cave



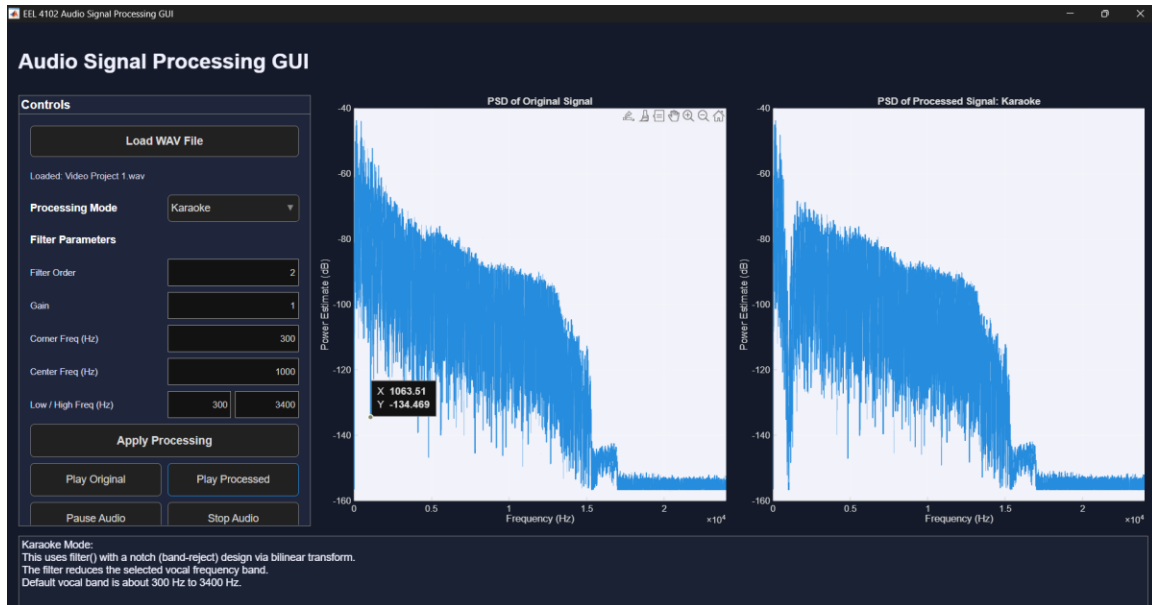
3.2 Cathedral



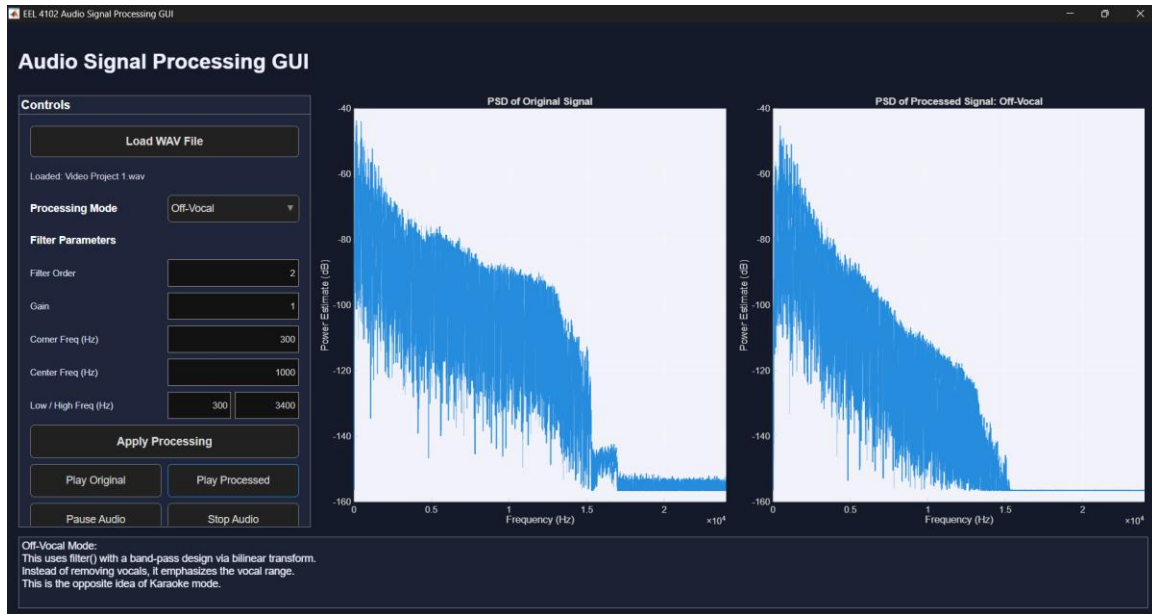
3.3 stadium



3.4 Karaoke



3.5 Off Vocal



4. Discussion of Results

4.1 Cave, Cathedral, and Stadium

When convolution is applied with a room impulse response, the primary perceptual effect is the addition of reverberation — the audio sounds like it was recorded inside a physical space. In the PSD plots, the frequency content of the original and processed signals looks broadly similar because convolution with a noise-based impulse response adds energy relatively evenly across the spectrum rather than cutting specific bands. However, slight differences in the PSD are visible, particularly at higher frequencies, because the

exponential decay envelope in the impulse response attenuates high-frequency reflections more than low-frequency ones.

The Cathedral mode produces the longest reverb tail and the most audible change. The PSD of the processed signal may show a slight elevation at low and mid frequencies compared to the original because the slow decay keeps low-frequency energy present for longer. The Cave mode sounds noticeably more enclosed, with quicker reflections that make the audio feel tight. The Stadium mode is perceptually distinct from Cave and Cathedral because the wider reflection delays produce echo that can be heard as a separate event rather than a smooth reverb.

An important technical note is that `conv()` with the 'full' option produces an output longer than the input. This is expected — the extra length is the reverb tail, where the room reflections continue after the source sound has ended. The output is normalized before playback and PSD display.

4.2 Karaoke Mode

The Karaoke mode applies a notch (band-reject) filter using the `lsim()` function with state-space matrices. The filter is based on the analog transfer function $H(s) = (s^2 + \omega_0^2) / (s^2 + Bw \cdot s + \omega_0^2)$, which places a zero at the center frequency ω_0 , driving signal power to zero at that point. The default settings target the vocal frequency range of approximately 300–3400 Hz. The PSD of the processed signal should show a noticeable drop in energy within that band compared to the original.

However, complete vocal removal is rarely achievable because vocals overlap in frequency with many instruments such as guitar, piano, and synths. The notch filter cannot distinguish between a vocal and an instrument at the same frequency — it removes all signal content in the rejected band equally. Additionally, increasing the Filter Order cascades the filter multiple times, which deepens and widens the notch but also removes more of the surrounding musical content. These are fundamental limitations of frequency-domain filtering, not errors in the implementation.

4.3 Off-Vocal Mode

Off-Vocal mode applies a band-pass filter using the same `lsim()` approach with the transfer function $H(s) = (Bw \cdot s) / (s^2 + Bw \cdot s + \omega_0^2)$. This is the mathematical complement of the Karaoke notch — it keeps only the frequencies within the selected band and suppresses everything outside it. The A and B state-space matrices are identical to Karaoke mode because both filters share the same denominator, meaning the same resonant structure. Only the C and D matrices differ, reflecting the different numerator that defines what is passed versus rejected. The PSD of the processed signal shows a strong concentration of energy within the vocal band and significantly reduced energy at all other frequencies. The audio sounds thin because the bass, high-frequency content, and out-of-band instruments have been filtered out, leaving only the selected range.

4.4 PSD as a Diagnostic Tool

The side-by-side PSD comparison is the most informative part of the GUI for understanding what each mode actually does to the signal. The room modes (Cave, Cathedral, Stadium) primarily affect the time-domain behavior by adding reflections; their PSDs look similar to the original because the effect is a temporal spreading of energy rather than a spectral reshaping. The filter modes (Karaoke, Off-Vocal) produce clearly visible changes in the PSD because they actively remove or retain specific frequency bands. This distinction — time-domain effects vs. frequency-domain effects — is one of the core concepts the assignment is designed to illustrate.